

VIET NAM NATIONAL UNIVERSITY HO CHI MINH CITY

INTERNATIONAL UNIVERSITY

Fundamental Concepts of Data Security

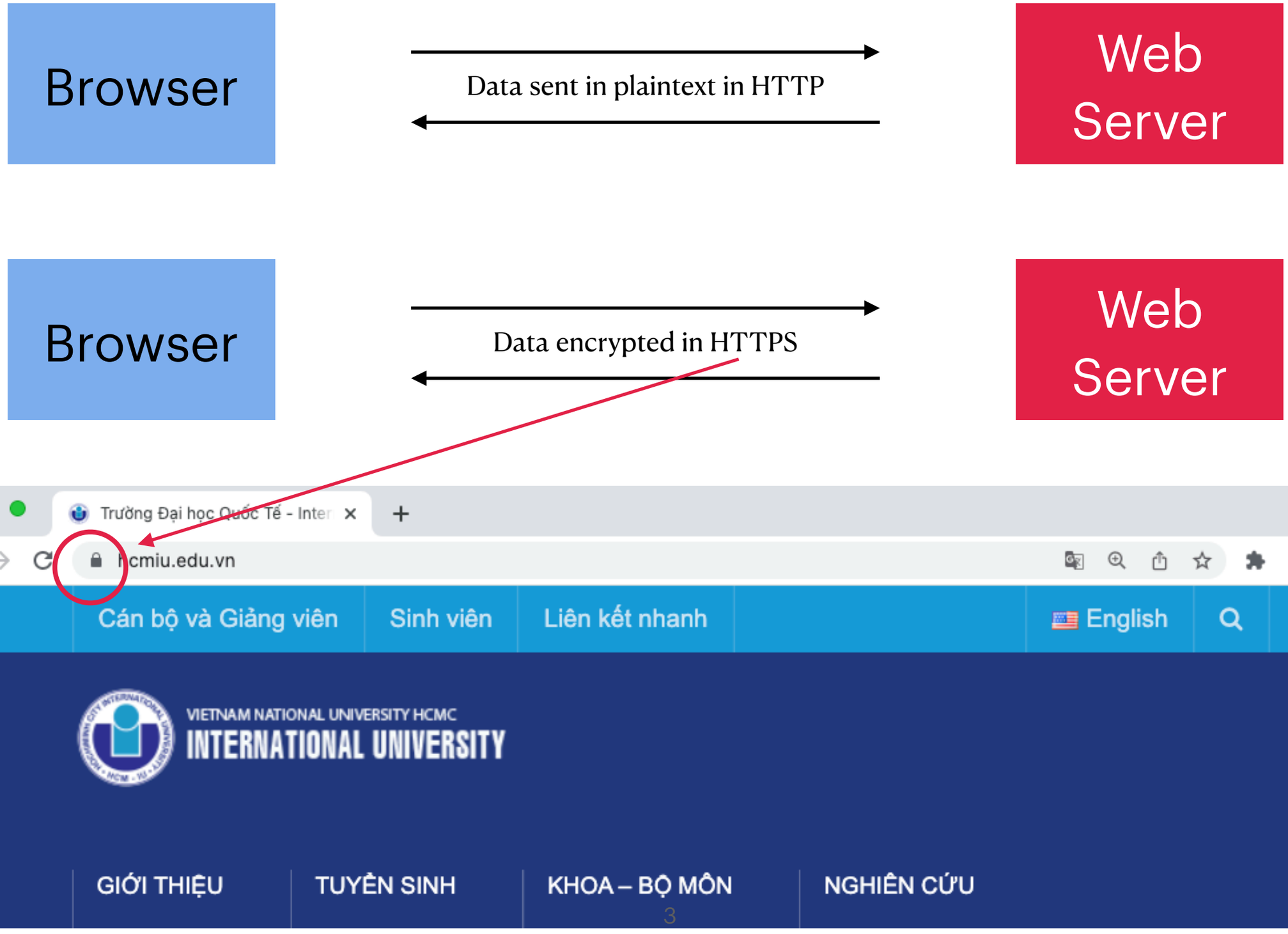
Applied Cryptography

Le Hai Duong, PhD. (lhduong@hcmiu.edu.vn)

Objectives

- Overview the uses of cryptography
- Introduce concept of encryption, message authentication, and public key system

How to keep your data secret?



Encryptions

- Symmetric encryption (one key)
- Asymmetric encryption (two keys)

Symmetric Encryption

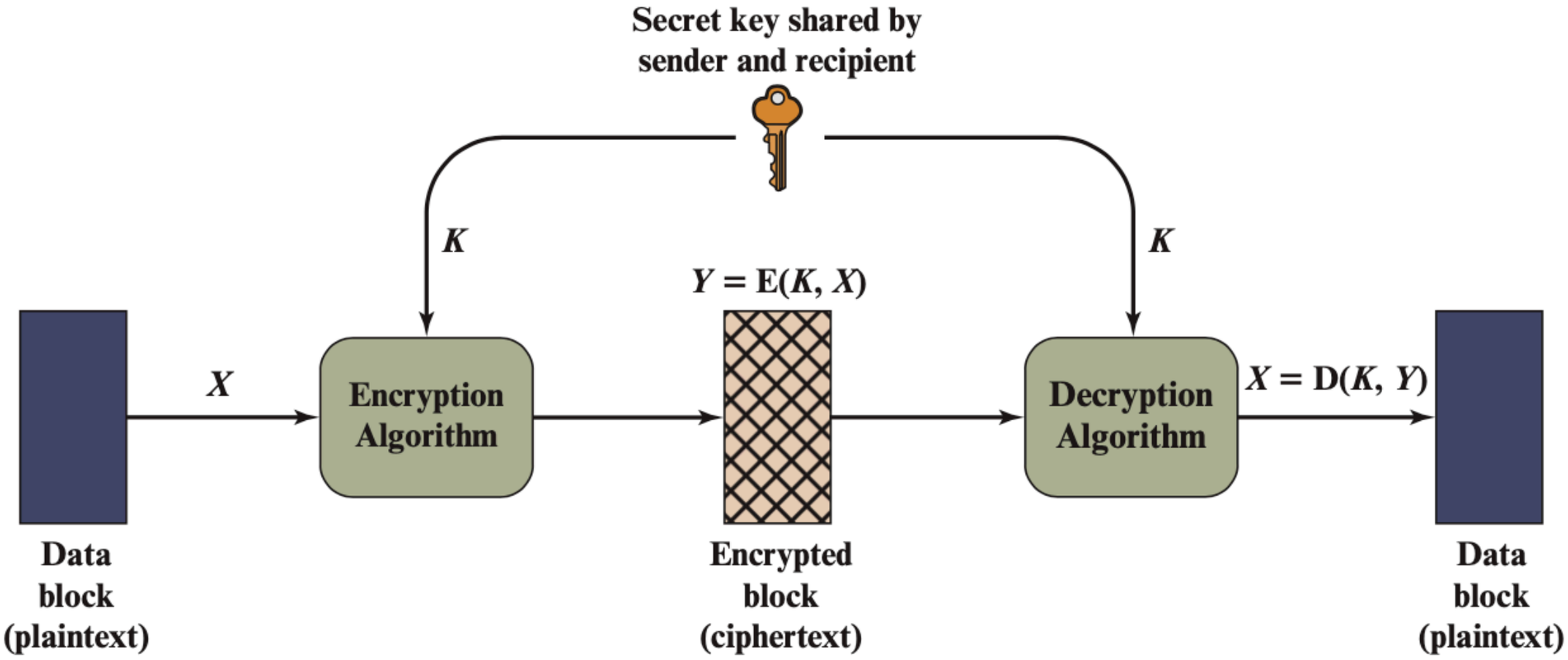
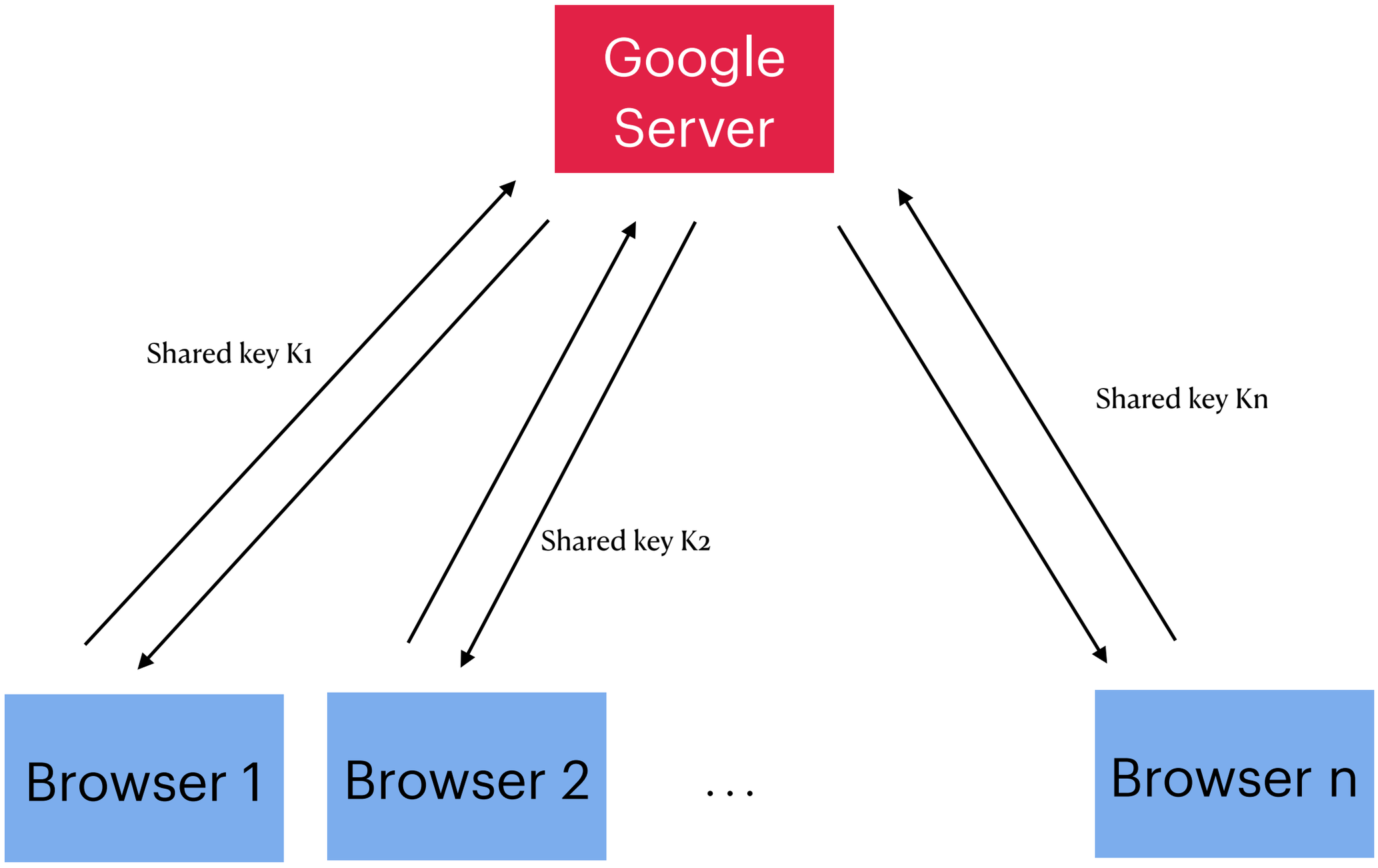


Figure 3.1 Simplified Model of Symmetric Encryption

Troubles when using symmetric key

- How to share the key??? (**Key Exchange**)
    - Share over the phone
    - Send by... “Grabkey”
- Timing problem

How many keys should Google server keep?

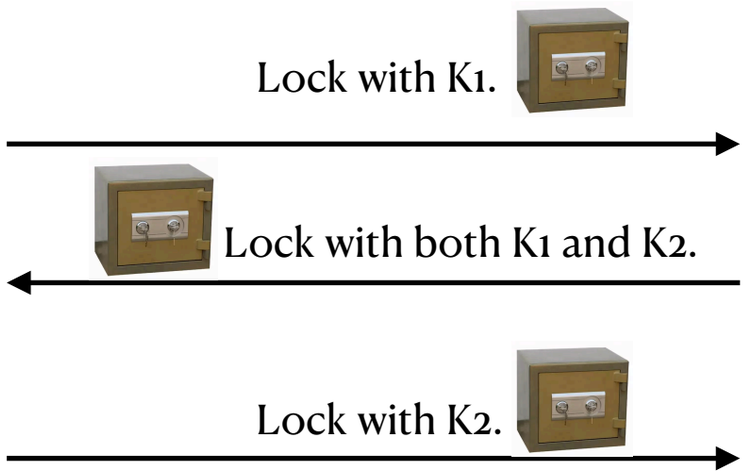


How about 2 keys system?



Alice

(1) Put the secret message in the box
(2) Lock the box with key K1.



Bob

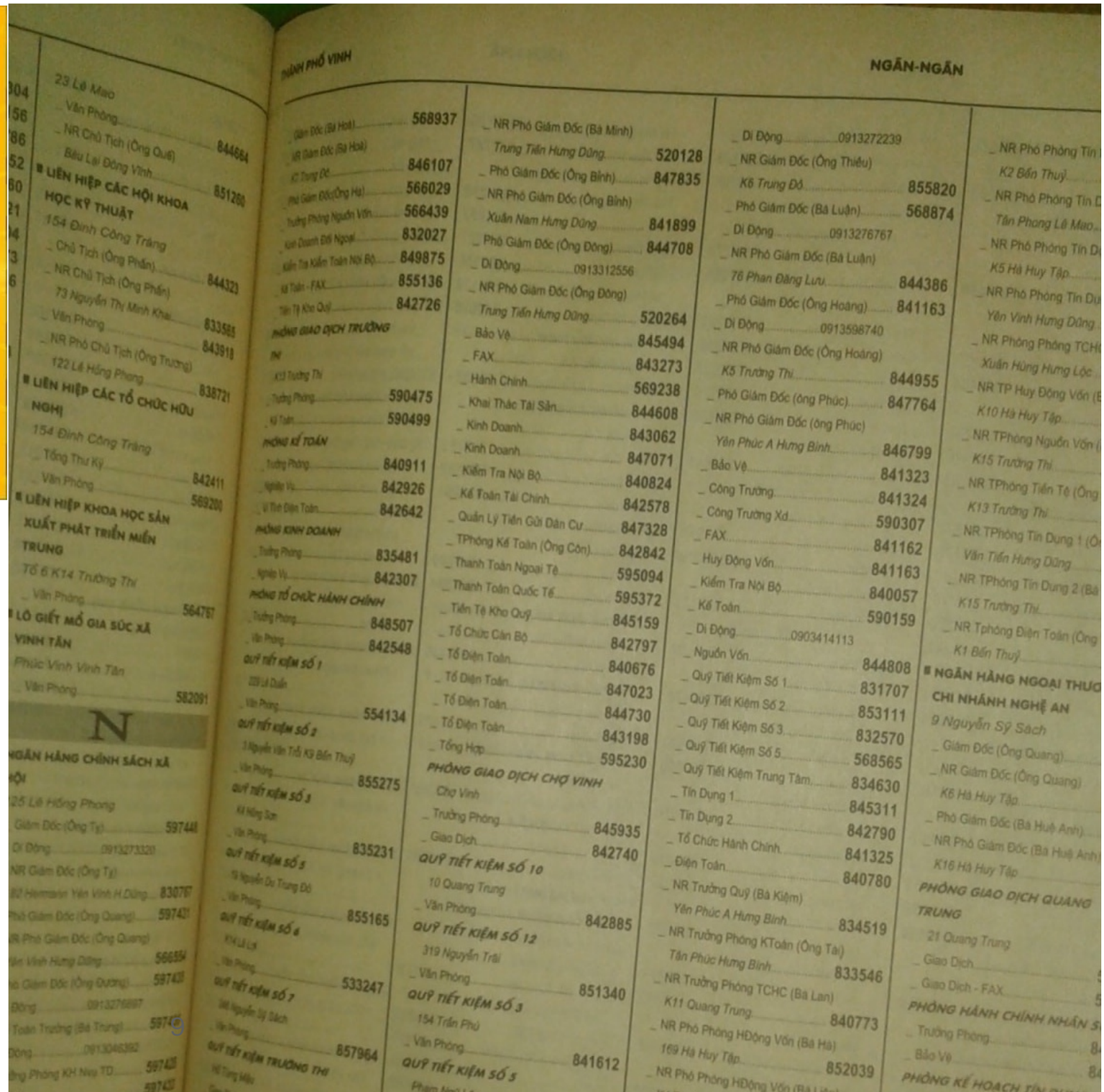
(3) Lock the box with key K2.

(4) Unlock with K1.

(5) Unlock with K2.
(6) Get the message

The Yellow Pages Idea





Alice

Bob

Alice want to send secret message to Bob?

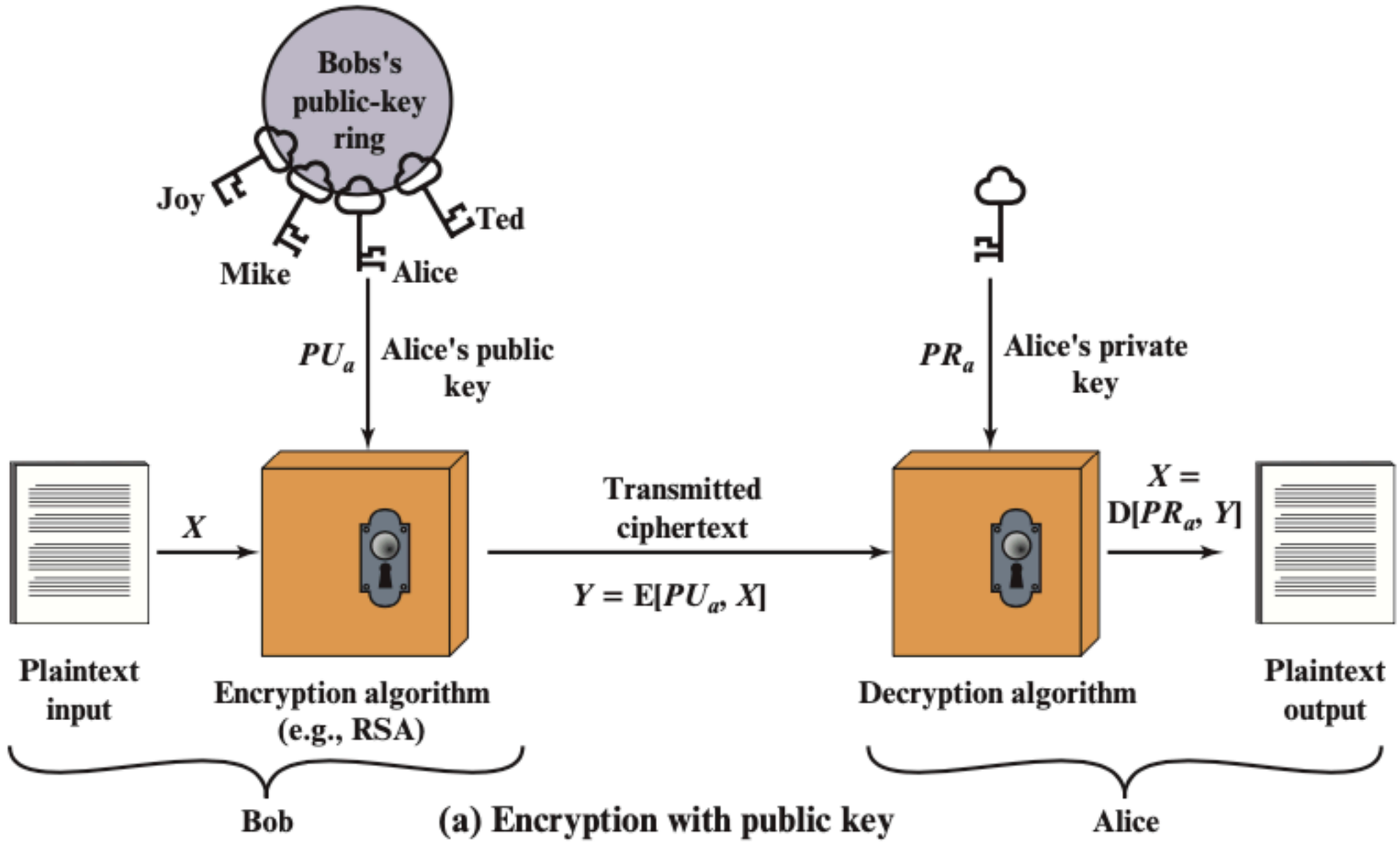
(1) Look up "Yellow Pages" for the Bob's **public key PU_B**

(2) Encrypt message with that key and send the ciphertext **C**

Ciphertext **C**

(3) Bob decrypt the cipher text C with his **private (secret) key PR_B**

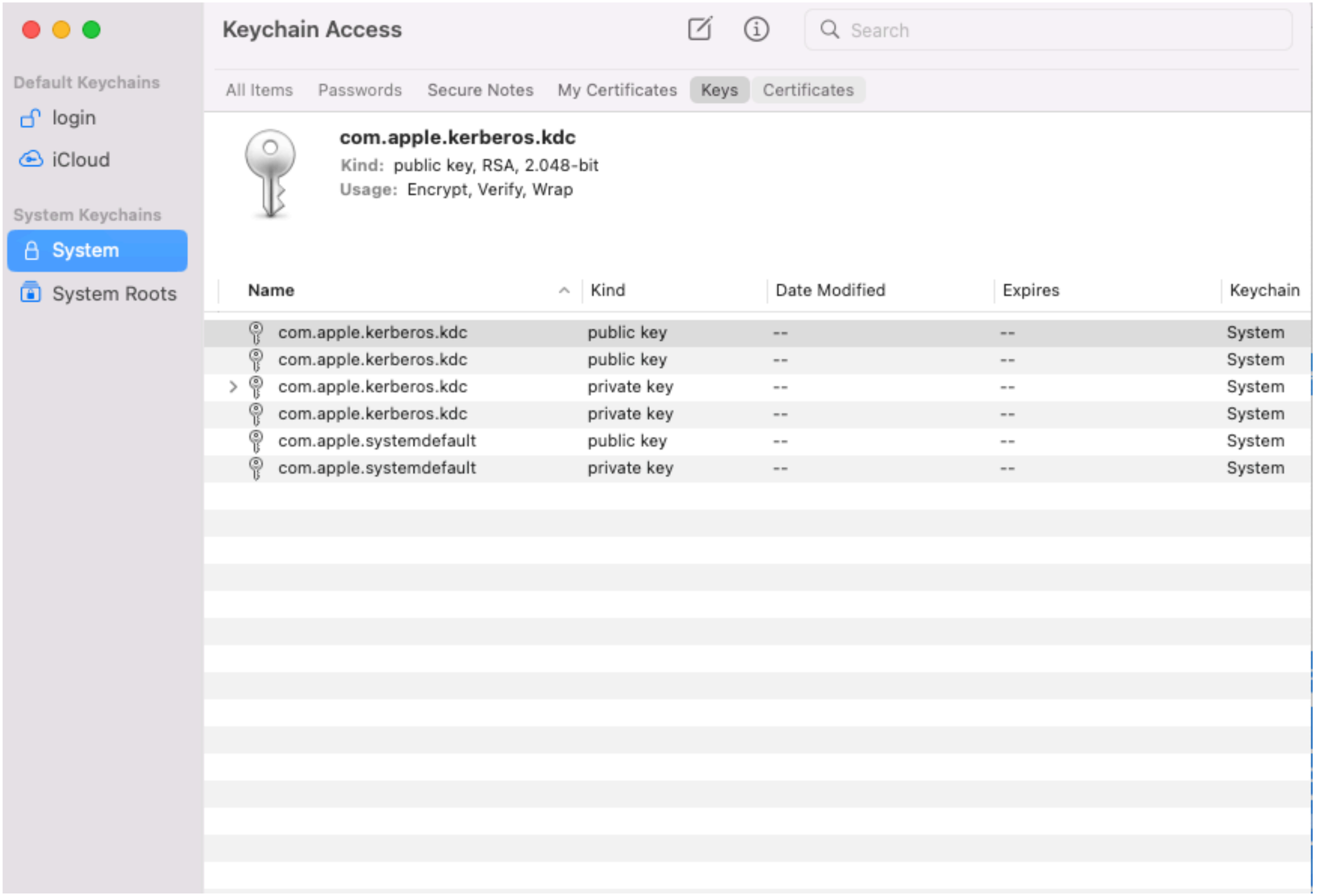
Public Key System



(a) Encryption with public key

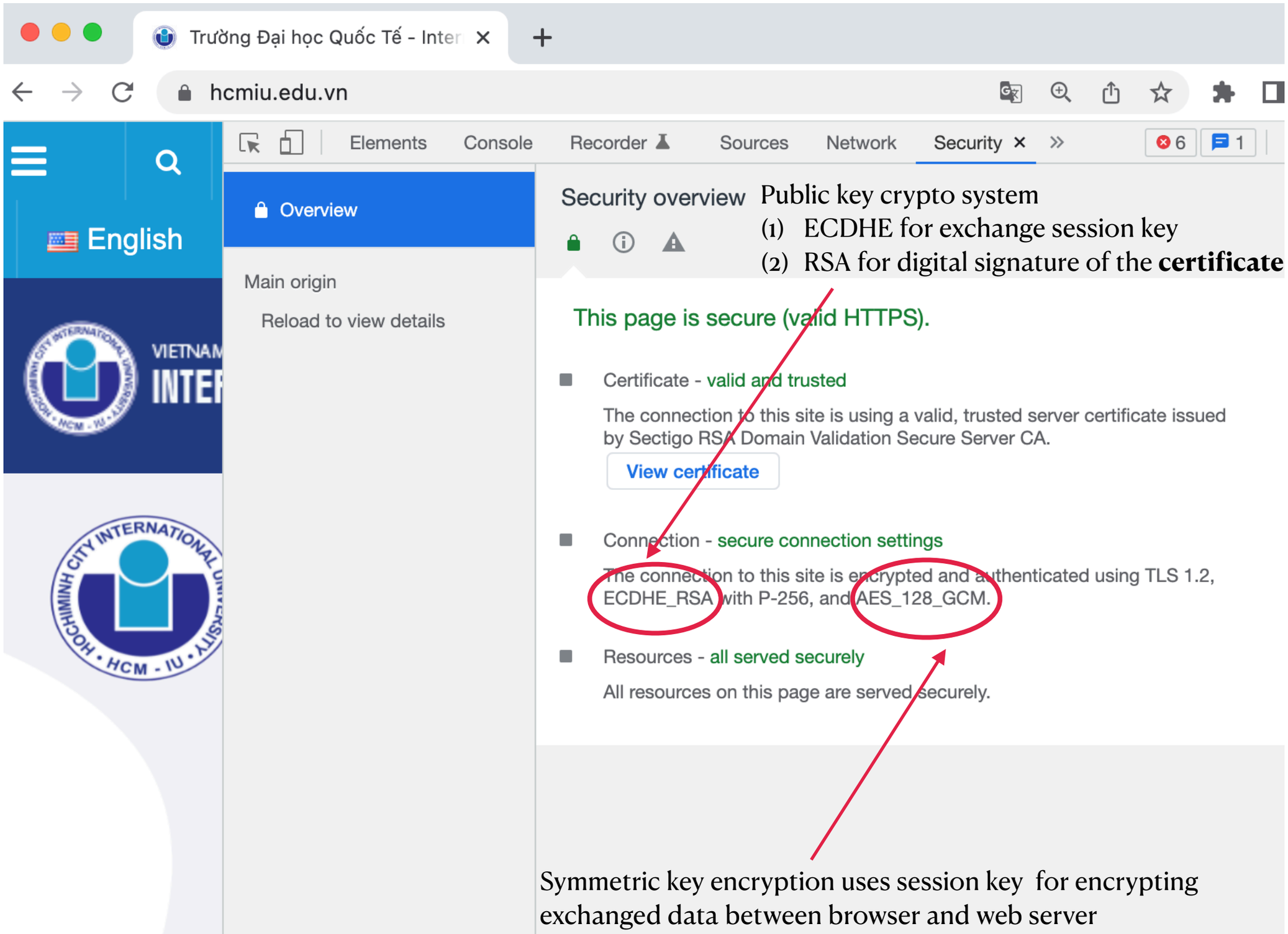
Public key encryption use public key to encrypt!!!

Key ring on macOS



Use public key crypto to encrypt web traffic/data encryption???

- NOPE
  - Public key algorithm runs... too slow compared to symmetric key encryption
  - Solution: Combine both asymmetric key + symmetric key
    - Asymmetric key system —> exchange session key
    - Symmetric key encryption using session key


Trường Đại học Quốc Tế - Inter
hcmiu.edu.vn
English
VIETNAM
INTER
Elements
Console
Recorder
Sources
Network
Security
6
1
Overview
Main origin
Reload to view details
Security overview
Public key crypto system
(1) ECDHE for exchange session key
(2) RSA for digital signature of the certificate
This page is secure (valid HTTPS).
Certificate - valid and trusted
The connection to this site is using a valid, trusted server certificate issued by Sectigo RSA Domain Validation Secure Server CA.
View certificate
Connection - secure connection settings
The connection to this site is encrypted and authenticated using TLS 1.2, ECDHE_RSA with P-256, and AES_128_GCM.
Resources - all served securely
All resources on this page are served securely.
Symmetric key encryption uses session key for encrypting exchanged data between browser and web server

Example of Using Public Key CryptoSystem for Key Exchange

Browser (B)

Web Server (WS)

Browser wants to exchange a session key $\mathbf{K_S}$

(1) Encrypt session key $\mathbf{K_S}$ with public key $\mathbf{PU_{WS}}$ of the web server

$E(\mathbf{PU_{WS}}, \mathbf{K_S})$

K_S

(2) Web Server uses its private key $\mathbf{PR_{WS}}$ to decrypt and obtains the session key $\mathbf{K_S}$

$E(\mathbf{K_S}, \text{data_from_browser})$

$E(\mathbf{K_S}, \text{data_from_server})$

How to distribute the public key???

(1) Post public key on the Key Server, e.g. pgp.mit.edu

pgp.mit.edu/pks/lookup?search=duonghaile%40gmail.com&op=index

Search results for 'gmail duonghaile com'

```
Type bits/keyID     Date       User ID
pub  2048R/3D492272 2019-03-01 Le Hai Duong (Sign my mail) <duonghaile@gmail.com>
pub  1024R/435B82BD 2017-11-13 Le Hai Duong <duonghaile@gmail.com>
pub  4096R/B78ED08A 2016-05-1
pub  2048R/3433B1CF 2015-07-0
pub  2048R/755DC0E0 2015-02-0
```

Call me to confirm my public key's fingerprint

pgp.mit.edu/pks/lookup?op=get&search=0x29[illegible]443D492272

Public Key Server -- Get "0x290d74443d492272 "

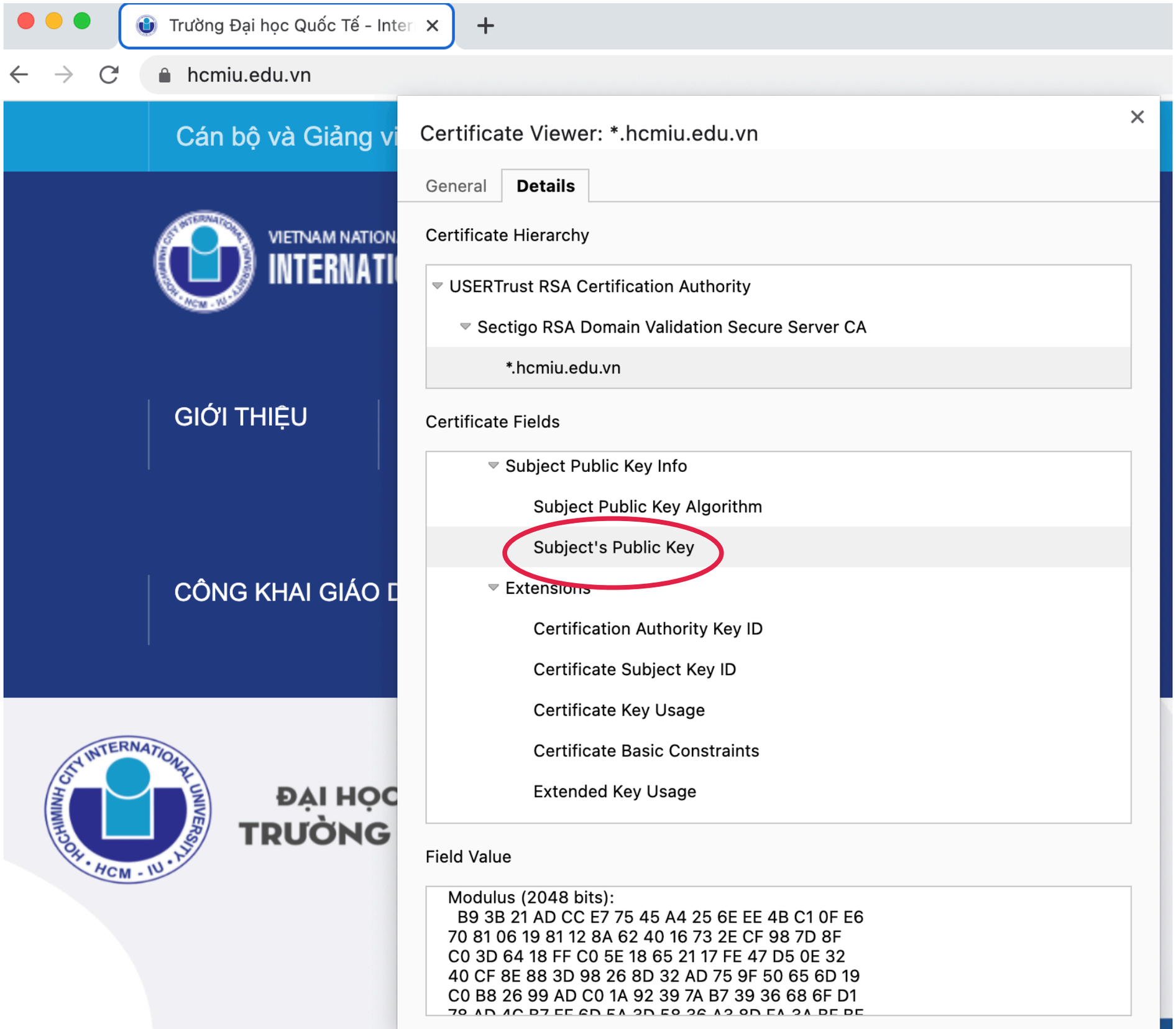
FINGERPRINT of the public key = Hash of the public key

```
-----BEGIN PGP PUBLIC KEY BLOCK-----
Version: SKS 1.1.6
Comment: Hostname: pgp.mit.edu

mQENBFx4f/kBCADTxW/LzH9Q2+2x/aesDGP/z1hGOOMWZ9hzaDUraF1AnLDYa1szYpcis3US
3BRTVqwZ9mY+CWsZ2gc/VHo9ZBlHhCn4/u2l83VOoPyoxrbNLZZbpBIdCUODYpYqLWPajCa8
Zo6cHPS/EFIyvDXFUP1tajE53v0KFUJjpz+drIpH0sCesshd4aT+2FJV0dG5NGA2ic9HTMnn
KoApMvP9NRcGhnO6utZpQVxM/C0V3rpsU1eI3V95VDM7DWA32qRD13E7FTj4HBZH3sSWAB78
9TVN/ZshfYi6fKu4VwhGYcXoXLxv2fQCwW8fhcCuItEroRgYbdVwQPGCEzUZCsr+iuLpABEB
AAG0MkxlIEhhaSBEdW9uZyAoU2lnbiBteSBtYWlsKSA8ZHVvbmdoYWlsZUBnbWFpbC5jb20+
iQEiBBABAgAMBQJcikJLBQMAEnUAAAoJEJcQuJvKV618DfAH/iV5boi3tIG1KLOOPxyJDN4A
Uc1HjyzC0p5qM+VEa09mfTMdiUXuYKRk1qzMvAzufy3ySAkxjvgAQfcI04LyajRVGUn3jcBJ
```

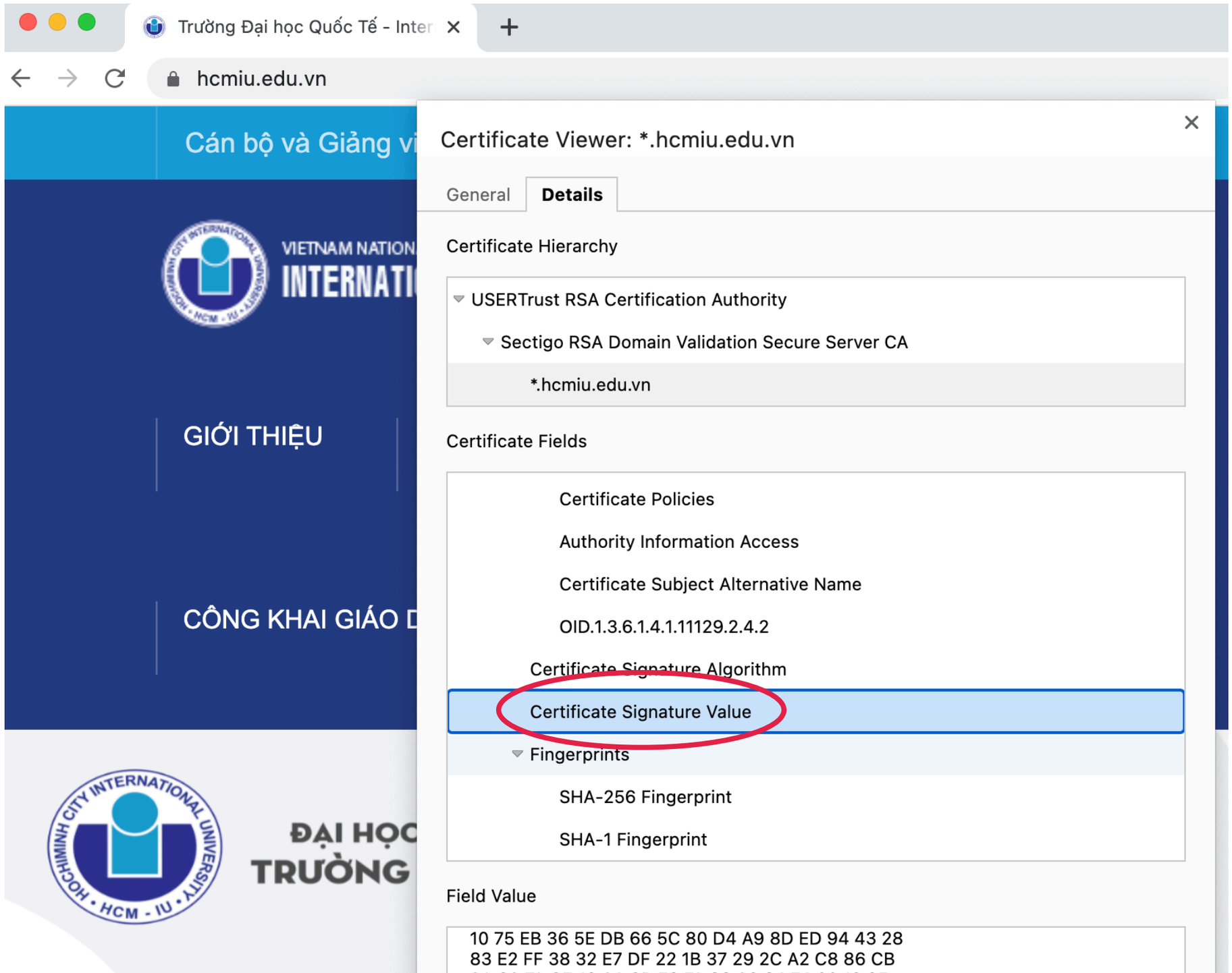
How to distribute the public key???

(2) Web server sends public key inside a **certificate**



How do we trust the public key inside a certificate??

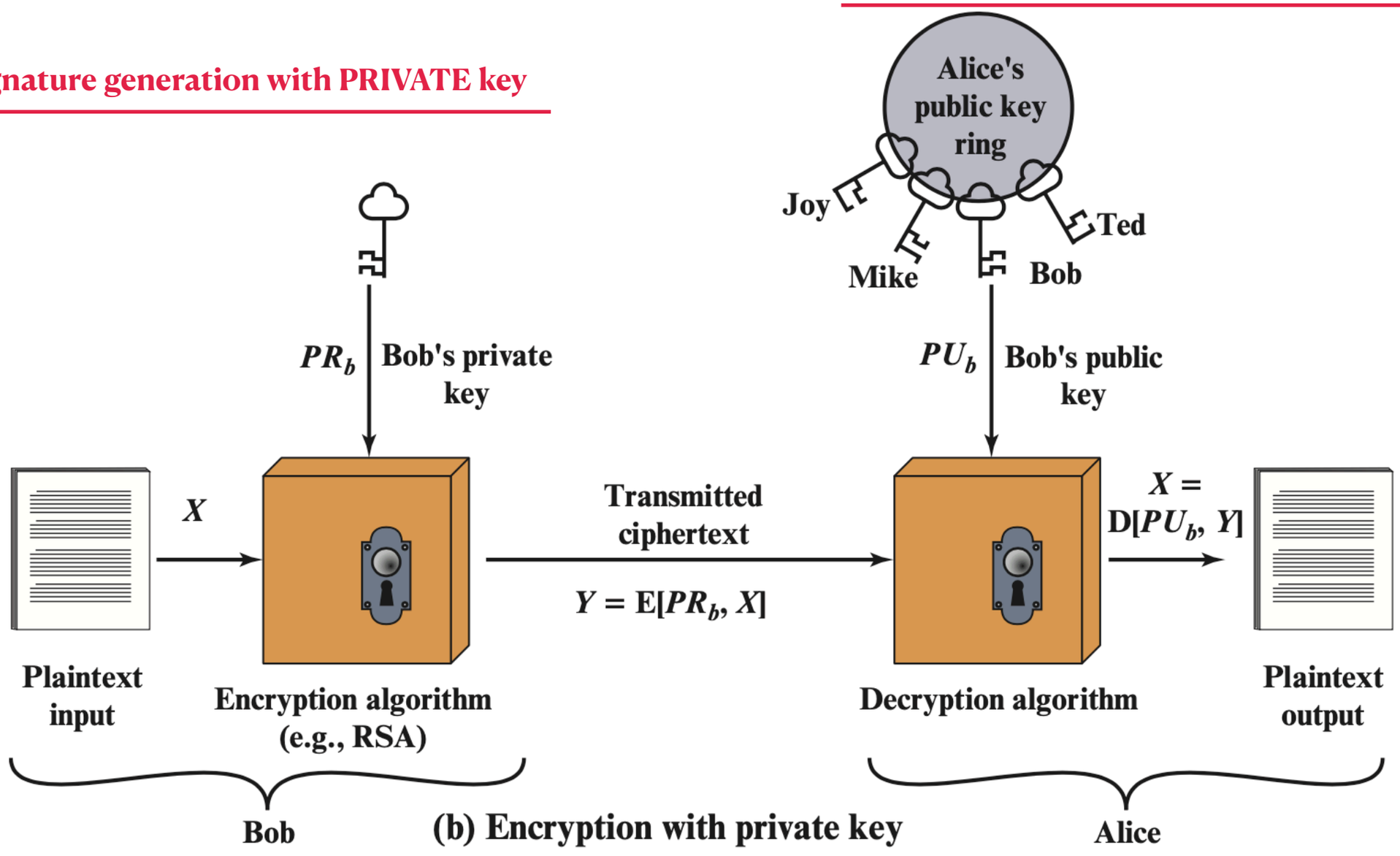
- Certificate containing the public key is SIGNED (digitally) by an issuer (a trusted party)



Digital Signature

Signature generation with PRIVATE key

Signature verification with PUBLIC key



(b) Encryption with private key

Encrypt using private key —-> digitally sign!!!

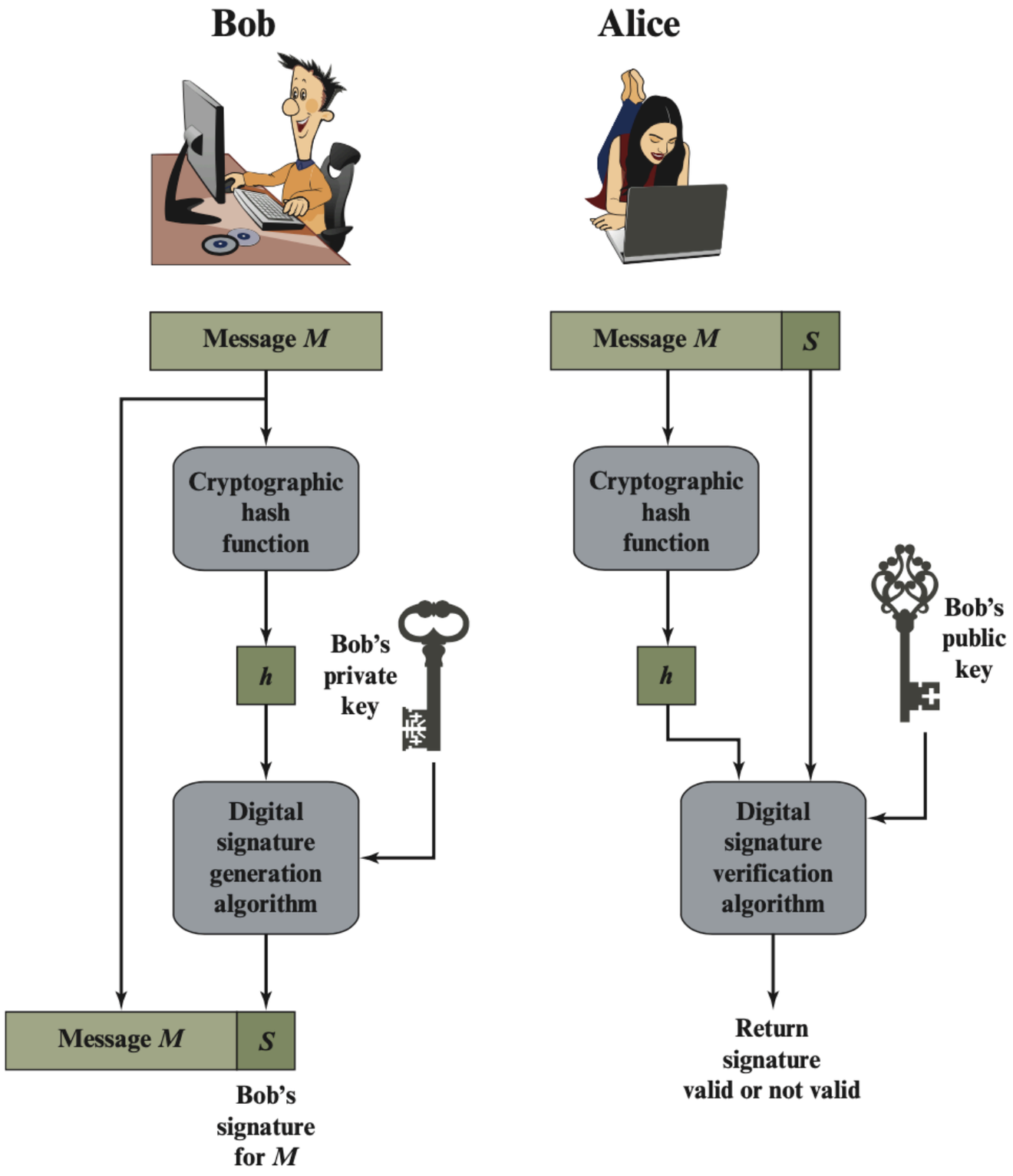
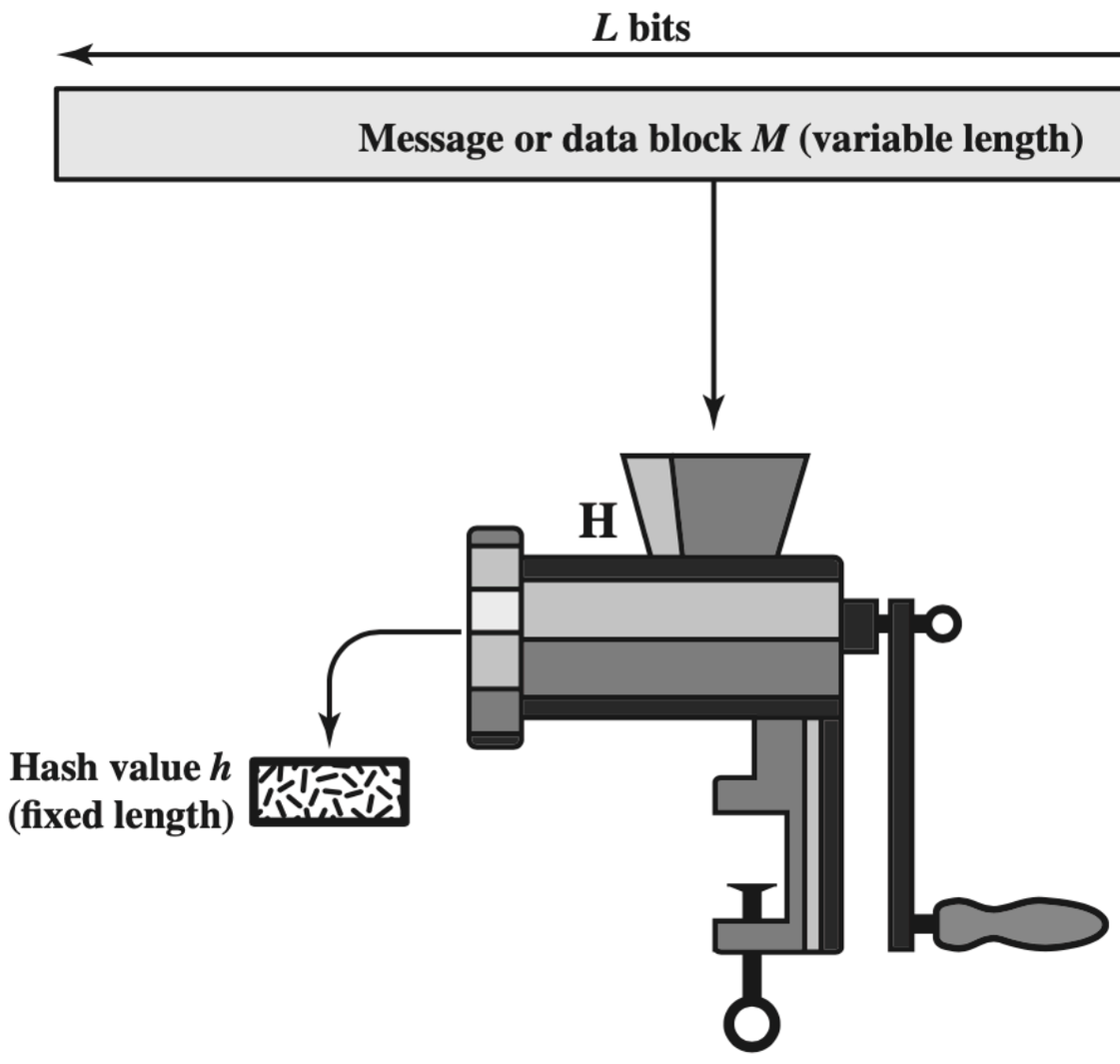


Figure 13.1 Simplified Depiction of Essential Elements of Digital Signature Process

Hash Function



P, L = padding plus length field

Figure 2.4 Cryptographic Hash Function; $h = H(M)$

- It's infeasible find two inputs producing the same hash value, i.e. a hash collision.
- Given M1 and h = H(M1), it's infeasible to find M2 that H(M2) = H(M1) = h —> can't create fake message with the same hash value

Exercises

(1) Design a system for encrypting files on your computer using combination of public key encryption and symmetric key encryption.

(2) Generate secret key for symmetric key encryption using a password and hash function?

(3) How to secure the private key on your computer?